

分部積分法

8-1 Integration by Parts

師大工教一

Integration by Parts: Indefinite Integrals

$$\frac{d}{dx}(u(x)v(x)) = u'(x)v(x) + u(x)v'(x)$$

$$\Rightarrow \int \frac{d}{dx}(u(x)v(x)) dx = \int [u'(x)v(x) + u(x)v'(x)] dx$$

$$\Rightarrow \int \frac{d}{dx}(u(x)v(x)) dx = \int u'(x)v(x) dx + \int u(x)v'(x) dx$$

$$\Rightarrow \int u(x)v'(x) dx = \int \frac{d}{dx}(u(x)v(x)) dx - \int u'(x)v(x) dx$$

Formula: $\int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$

Differential Version: $\int u dv = uv - \int v du$

Ex1(p456) Find $\int x \cos x dx$.

$$= x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x + C$$

Note: What if choose u,v wrong?

Ex2(p457) Evaluate $\int \ln x \, dx$.

$$\begin{aligned} \text{orig} &= x \ln x - \int \frac{1}{x} \cdot x \, dx \\ &= x \ln x - x + C \end{aligned}$$

$$\begin{array}{r} \ln x \quad | \\ \frac{1}{x} \quad \int \quad x \end{array}$$

Ex3(p457) Find $\int x^2 e^x dx$.

origm = $x^2 e^x - 2x e^x + 2e^x - \int 0 dx$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

$$\begin{array}{r} x^2 e^x \\ \swarrow - \\ x e^x \\ \swarrow - \\ 2 e^x \\ \swarrow - \\ 0 \end{array}$$

Ex4(p458) Evaluate $\int e^x \cos x dx$.

$$\begin{aligned} \text{origm} &= e^x \cos x + e^x \sin x - \int e^x \cos x dx \\ 2 \int e^x \cos x dx &= e^x \cos x + e^x \sin x + C \\ \int e^x \cos x dx &= \frac{e^x \cos x + e^x \sin x + C}{2} \end{aligned}$$

Diagram illustrating the integration by parts process:

cos x	/	+ e ^x
-sin x	/	e ^x
-cos x	/	- e ^x
+ sin x	/	e ^x

$\int e^x \cos x dx$ or $\int e^{ax} \sin bx dx$
一定要做两次

Integration by Parts: Definite Integrals

$$\int_a^b f(x)g'(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)dx$$

Ex6(p459) Find the area of the region bounded by the curve $y = xe^{-x}$ and the x -axis from $x = 0$ to $x = 4$.

$$A = \int_0^4 xe^{-x} dx$$

$$= -xe^{-x}\Big|_0^4 + \int_0^4 e^{-x} dx$$

$$= -4e^{-4} + (-e^{-x}\Big|_0^4)$$

$$= -4e^{-4} - e^{-4} + 1$$

$$= -5e^{-4} + 1$$

✖

$$\begin{array}{l} x \quad e^{-x} \\ \swarrow + \\ \int -e^{-x} \end{array}$$

HW8-1

- HW:1,4,5,8,15,21,35,45,58.

$$\text{Ex}(102, 1(7)) \int x^2 e^{-x} dx = -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} - \int 0 dx$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + C$$

$$\begin{array}{r} x^2 \diagup e^{-x} \\ 2x \diagdown -e^{-x} \\ 2 \diagdown -e^{-x} \\ 0 \diagdown -e^{-x} \end{array}$$

$$\text{Ex}(102, 1(8)) \int \sin \sqrt{x} dx = \int \sin u \cdot 2u du$$

$$u = \sqrt{x}$$

$$x = u^2$$

$$dx = 2u du$$

$$= 2 \int u \sin u du$$

$$= 2 \left(-u \cos u + \int \cos u du \right)$$

$$= -2u \cos u + 2 \sin u + C$$

$$= -2\sqrt{x} \cos \sqrt{x} + 2 \sin \sqrt{x} + C$$

$$\begin{array}{r} u \sin u \\ \diagdown -\cos u \end{array}$$

$$\text{Ex}(103, 1(f)) \int \ln x dx$$

$$\text{Ex}(103, 1(g)) \int \frac{\ln x}{x} dx = \int u du$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$= \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$$

8-2 Trigonometric Integrals

師大工教一

Type 1-1: $\int \sin^m x \cdot \cos^n x dx$, at least one of m, n is odd.

Ex1(p463) Evaluate $\int \sin^3 x \cos^2 x dx$.

$$\text{origin} = \int \sin^2 x \cos^2 x (\sin x dx)$$

$$\text{Let } u = \cos x$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$= \int (1 - u^2) u^2 (-du)$$

$$= \int (u^4 - u^2) du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

Ex2(p464) Find $\int \cos^5 x \, dx$.

$$= \int \cos^4 x (\cos x \, dx)$$

$$\text{Let } u = \sin x \\ du = \cos x \, dx$$

$$= \int (1 - u^2)^2 \, du$$

$$= \int (1 - 2u^2 + u^4) \, du$$

$$= u - \frac{2u^3}{3} + \frac{u^5}{5} + C$$

$$= \sin x - \frac{2\sin^3 x}{3} + \frac{\sin^5 x}{5} + C$$

$$= \frac{\sin^5 x}{5} - \frac{2\sin^3 x}{3} + \sin x + C$$

★ Type 1-2: $\int \sin^m x \cdot \cos^n x dx$, both of m, n are even.

Ex3(p464) Evaluate $\int \sin^2 x \cos^4 x dx$.

$$= \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$= \frac{1}{8} \int (1 - \cos 2x)(1 + \cos 2x)^2 dx$$

$$= \frac{1}{8} \int (1 - \cos^2 2x)(1 + \cos 2x) dx$$

$$= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx$$

$$= \frac{1}{8} \left(x + \frac{1}{2} \sin 2x - \frac{1}{2} x - \frac{1}{8} \sin 4x - \frac{1}{2} \sin 2x + \frac{1}{6} \sin^3 2x \right) + C$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - \sin^2 x - \sin^2 x \\ &= 1 - 2\sin^2 x \end{aligned}$$

$$\begin{aligned} 2\sin^2 x &= 1 - \cos 2x \\ \sin^2 x &= \frac{1 - \cos 2x}{2} \end{aligned}$$

$$\begin{aligned} \int \cos^2 x dx \\ = \int \frac{1 + \cos 2x}{2} dx = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + C \end{aligned}$$

Ex4(p465) Evaluate $\int_0^{\frac{\pi}{4}} \sqrt{1 + \cos 4x} \, dx$.

$$= \int_0^{\frac{\pi}{4}} \sqrt{2\cos^2 2x} \, dx$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{2} \cos 2x \, dx$$

$$= \sqrt{2} \int_0^{\frac{\pi}{4}} |\cos 2x| \, dx$$

$$= \sqrt{2} \left(\frac{1}{2} \sin 2x \right) \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{\sqrt{2}} (\sin \frac{\pi}{2} - \sin 0)$$

$$= \frac{1}{\sqrt{2}}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$2\cos^2 x = 1 + \cos 2x$$

Type 2-1: $\int \tan^m x \cdot \sec^n x dx$, n is even.

Ex7(p466) Evaluate $\int \tan^4 x \sec^4 x dx$.

$$\text{Let } u = \tan x$$

$$du = \sec^2 x dx$$

$$\sec^2 x = 1 + \tan^2 x$$

$$= \int u^4 (u^2 + 1) du$$

$$= \int (u^6 + u^4) du$$

$$= \frac{u^7}{7} + \frac{u^5}{5} + C$$

$$= \frac{\tan^7 x}{7} + \frac{\tan^5 x}{5} + C$$

Type 2-2: $\int \tan^m x \cdot \sec^n x dx$, m is odd.

Ex(105 年分部, #1(3)) Find $\int \sec^5 x \tan^3 x dx$. $= \int \sec^4 x \tan^2 x (\sec x \tan x dx)$

$$u = \sec x$$

$$du = \sec x \tan x dx$$

$$\tan^2 x = \sec^2 x - 1$$

$$= \int u^4 (u^2 - 1) du$$

$$= \int u^6 - u^4 du$$

$$= \frac{u^7}{7} - \frac{u^5}{5} + C$$

$$= \frac{\sec^7 x}{7} - \frac{\sec^5 x}{5} + C$$



Ex6(p466) Find $\int \sec^3 x dx$.

$$\begin{array}{c} \sec x \quad \sec^2 x \\ \swarrow \quad \searrow \\ \sec x \tan x \quad \int \tan x \end{array}$$

$$= \sec x \tan x - \int \sec x \tan^2 x dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \tan x - \int (\sec^3 x - \sec x) dx$$

$$= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$\Rightarrow \int \sec^3 x dx = \sec x \tan x + \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C'$$

$$\int \sec x dx$$

$$= \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx$$

$$\text{Let } u = \sec x + \tan x$$

$$du = (\sec x \tan x + \sec^2 x) dx$$

$$= \sec x (\sec x + \tan x) dx$$

$$= \int \frac{1}{u} du = \ln |u| + C$$

$$= \ln |\sec x + \tan x| + C$$

Using Product Identities

$$(a) \sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$(b) \sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$(c) \cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

Ex8(p467) Evaluate $\int \sin 3x \cos 5x \, dx$.

$$= \frac{1}{2} \int \sin(8x) + \sin(-2x) \, dx$$

$$= \frac{1}{2} \int (\sin 8x - \sin 2x) \, dx$$

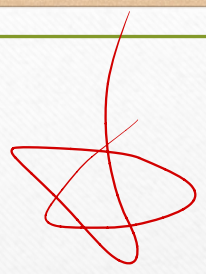
$$= \frac{1}{2} \left(-\frac{\cos 8x}{8} + \frac{1}{2} \cos 2x \right) + C$$

HW8-2

- HW: 3, 5, 11, 14, 19, 26, 42, 51

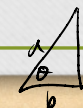
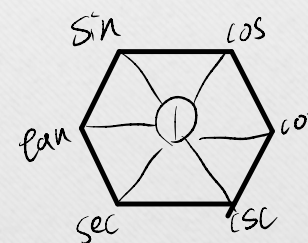
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$$\int \tan^3 x \sec^4 x dx$$



8-3 Trigonometric Substitution

師大工教一



$$\begin{aligned}\sin &= \frac{c}{a} & \csc &= \frac{a}{c} \\ \cos &= \frac{b}{a} & \sec &= \frac{a}{b} \\ \tan &= \frac{c}{b} & \cot &= \frac{b}{c}\end{aligned}$$

$$\begin{aligned}\sin^2 + \cos^2 &= 1 \\ \tan^2 + 1 &= \sec^2 \\ \cot^2 + 1 &= \csc^2\end{aligned}$$

$$\frac{\sin}{\cos} = \tan$$

$$\frac{\cos}{\sin} = \cot$$

$$\frac{\sec}{\csc} = \tan$$

$$\frac{\csc}{\sec} = \cot$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

Type I: Integrals with $\sqrt{x^2 + a^2}$

Ex1(p470) Evaluate $\int \frac{1}{\sqrt{4+x^2}} dx$.

$$\text{Let } x = 2 \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$dx = 2 \sec^2 \theta d\theta$$

$$\sqrt{4+x^2}$$

$$= \sqrt{4+4\tan^2 \theta}$$

$$= 2\sqrt{1+\tan^2 \theta}$$

$$= 2\sqrt{\sec^2 \theta}$$

$$= 2|\sec \theta|$$

$$= \int \frac{1}{\cancel{2\sec \theta}} (2\sec^2 \theta d\theta) = \int \sec \theta d\theta$$

$$\left. \begin{array}{l} \tan \theta = \frac{x}{2} \quad \frac{\sqrt{4+x^2}}{2} \\ \sec \theta = \frac{\sqrt{4+x^2}}{2} \end{array} \right| = \ln |\sec \theta + \tan \theta| + C$$

$$= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C$$

Type II: Integrals with $\sqrt{a^2 - x^2}$

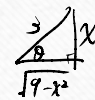
Ex3(p471) Evaluate $\int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{\cancel{3 \cos \theta}} \cdot \cancel{3 \cos \theta} d\theta = \int 9 \sin^2 \theta d\theta$

Let $x = 3 \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$dx = 3 \cos \theta d\theta$

$$\begin{aligned} & \sqrt{9-x^2} \\ &= \sqrt{9-9\sin^2 \theta} \\ &= 3\sqrt{\cos^2 \theta} \\ &= 3|\cos \theta| \end{aligned}$$

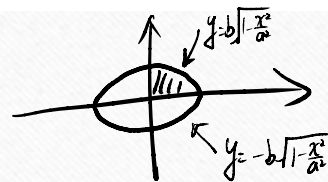
$x = 3 \sin \theta$
 $\sin \theta = \frac{x}{3} \quad \cos = \frac{\sqrt{9-x^2}}{3}$



$\theta = \sin^{-1}(\sin \theta) = \sin^{-1}(\frac{x}{3})$

$$\begin{aligned} &= 9 \int \frac{1-\cos 2\theta}{2} d\theta \quad \text{[Note: } \sin \theta \cos \theta \text{]} \\ &= \frac{9}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) + C \\ &= \frac{9}{2} \left(\sin^{-1}(\frac{x}{3}) - \frac{x\sqrt{9-x^2}}{9} \right) + C \end{aligned}$$

Ex Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\frac{y}{b} = \pm \sqrt{1 - \frac{x^2}{a^2}}$$

$$y = \pm b \sqrt{1 - \frac{x^2}{a^2}}$$

$$4 \int_0^a b \sqrt{1 - \frac{x^2}{a^2}} dx$$

$$= 4b \int_0^a \sqrt{1 - \frac{x^2}{a^2}} dx$$

$$\text{Let } x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$dx = a \cos \theta d\theta$$

$$\sqrt{1 - \frac{a^2 \sin^2 \theta}{a^2}}$$

$$= |\cos \theta|$$

$$= 4b \int_0^{\frac{\pi}{2}} a \cos^2 \theta d\theta$$

$$= 4ab \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= 4ab \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= 2ab \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{2}}$$

$$= 2ab \left(\frac{\pi}{2} + 0 - 0 - 0 \right)$$

$$= \pi ab$$

Type III: Integrals with $\sqrt{x^2 - a^2}$

Ex4(p471) Evaluate $\int \frac{dx}{\sqrt{25x^2 - 4}}, x > \frac{2}{5}.$ $= \int \frac{\cancel{\frac{2}{5}} \sec \theta \cancel{\tan \theta} d\theta}{\cancel{\frac{2}{5}} \tan \theta}$

Let $x = \frac{2}{5} \sec \theta, 0 \leq \theta < \frac{\pi}{2}$

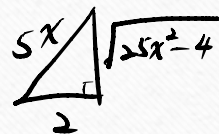
$dx = \frac{2}{5} \sec \theta \tan \theta d\theta$

$\sqrt{25x^2 - 4}$
 $= \sqrt{4 \sec^2 \theta - 4}$

$= 2 \sqrt{\sec^2 \theta - 1}$

$= 2 |\tan \theta|$

$x = \frac{2}{5} \sec \theta$
 $\sec \theta = \frac{5x}{2}$



$\tan \theta = \frac{\sqrt{25x^2 - 4}}{2}$

$= \frac{1}{5} \int \sec \theta d\theta$

$= \frac{1}{5} \ln |\sec \theta + \tan \theta| + C$

$= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2 - 4}}{2} \right| + C$

HW8-3

- HW:1,3,5,11,35,38,43,54,57

8-4 Integration of Rational Functions by Partial Fractions

用部分分式求有理函數的積分

師大工教一

Recall: Rational functions $f(x) = \frac{P(x)}{Q(x)}$, where P, Q are polynomials.

If $\deg(P(x)) \geq \deg(Q(x))$, use division and transfer $f(x)$ into a polynomial and a proper fraction. Now consider the proper fraction part.

For example: Q: Evaluate $\int \frac{x^3 - x}{x^2 + x + 1} dx$.

By polynomial long division, we have $\int \frac{x^3 - x}{x^2 + x + 1} dx = \int x - 1 + \frac{-x + 1}{x^2 + x + 1} dx$.

The Method of Partial Fractions: Express $R(x)$ as the sum of $\frac{A}{(ax+b)^i}$ or

$$\frac{Ax+B}{(ax^2+bx+c)^j}.$$

Case I $Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_nx + b_n)$

Ex1(p475) Evaluate $\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} dx$.

$$\begin{aligned} \text{Step I : } \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} &= \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3} \\ &= \frac{A(x+1)(x+3) + B(x-1)(x+3) + C(x-1)(x+1)}{(x-1)(x+1)(x+3)} \end{aligned}$$

$$\therefore x^2 + 4x + 1 = A(x+1)(x+3) + B(x-1)(x+3) + C(x-1)(x+1)$$

$$\text{plug } x=1 \Rightarrow 1+4+1 = A \cdot 2 \cdot 4 \Rightarrow 6 = 8A \quad A = \frac{3}{4}$$

$$\text{plug } x=-1 \Rightarrow 1-4+1 = -2 = B \cdot (-2) \cdot (2) \quad B = \frac{1}{2}$$

$$\text{plug } x=-3 \Rightarrow 9-12+1 = -2 = C \cdot (-4) \cdot (-2) \quad C = -\frac{1}{4}$$

$$\text{Step II : } \text{origin} = \int \frac{\frac{3}{4}}{x-1} + \frac{\frac{1}{2}}{x+1} - \frac{\frac{1}{4}}{x+3} dx = \frac{3}{4} \ln|x-1| + \frac{1}{2} \ln|x+1| - \frac{1}{4} \ln|x+3| + C$$

Case II $Q(x) = (a_1x + b_1)^{i_1} (a_2x + b_2)^{i_2} \cdots (a_nx + b_n)^{i_n}$

Ex2(p476) Evaluate $\int \frac{6x+7}{(x+2)^2} dx$.

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2} = \frac{A(x+2)+B}{(x+2)^2}$$

$$A=6 \quad B=-5$$

$$\text{origin} = \int \frac{6}{x+2} - \frac{5}{(x+2)^2} dx$$

$$= 6 \ln|x+2| + 5(x+2)^{-1} + C$$

Case III $Q(x)$ contains $ax^2 + bx + c$, where $b^2 - 4ac < 0$

Ex4(p477) Evaluate $\int \frac{-2x+4}{(x^2+1)(x-1)^2} dx$.

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$$

$$= \frac{A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2}{(x^2+1)(x-1)^2}$$

$$-2x+4 = A(x^3+x-x^2-1) + B(x^2+1) + (Cx+D)(x^2-2x+1)$$

$$\begin{aligned} A+C &= 0 \\ -A+B-2C+D &= 0 \end{aligned}$$

$$A+C-2D = -2 \quad D=1$$

$$-A+B+D = 2$$

$$4-2C=0 \quad \text{origin} = \int \left(\frac{-2}{(x-1)} + \frac{1}{(x-1)^2} + \frac{2x+1}{x^2+1} \right) dx$$

$$C=2$$

$$A=-2$$

$$B=1$$

$$= -2 \ln|x-1| - (x-1)^{-1} + \ln|x^2+1| + \tan^{-1}x + C$$

$$\int \frac{2x}{x^2+1} = \int \frac{1}{u} du = \ln|u| + C$$

Let $u=x^2+1$
 $du=2x dx$
 $= \ln|x^2+1| + C$

$$\int \frac{1}{x^2+1} dx = \tan^{-1}x + C$$

Ex5(p478) Evaluate $\int \frac{dx}{x(x^2+1)^2}$.

$$\frac{1}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$= \frac{A(x^2+1)^2 + (Bx+C)(x^2+1) + (Dx+E)(x)}{x(x^2+1)^2}$$

Ex6(p479) Find A, B, C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$$

HW8-4

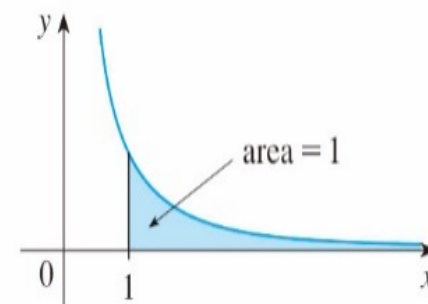
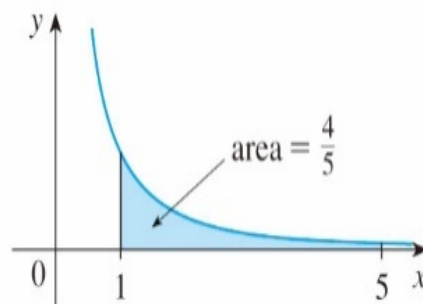
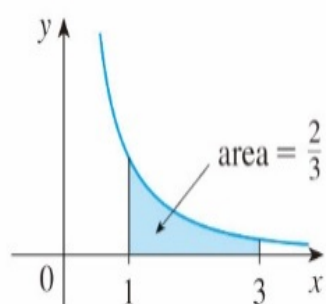
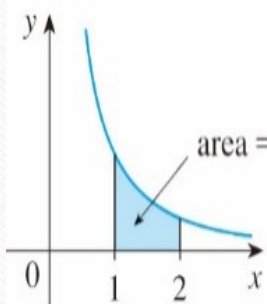
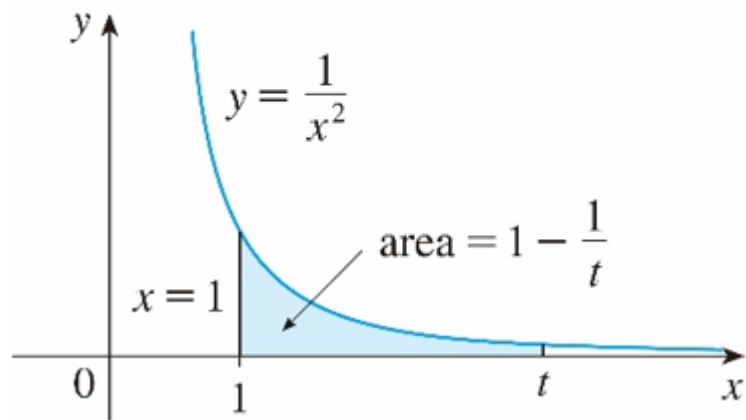
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- HW:10,11,17,25,29,47

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8-7 Improper Integrals

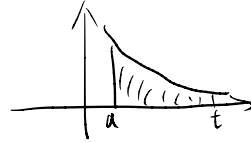
師大工教一

Q: $\int_1^{\infty} \frac{1}{x^2} dx = ?$

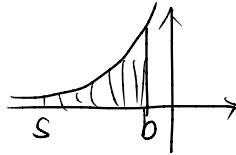


Definition of an Improper Integral of Type I

$$(a) \int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$



$$(b) \int_{-\infty}^b f(x) dx = \lim_{s \rightarrow -\infty} \int_s^b f(x) dx$$

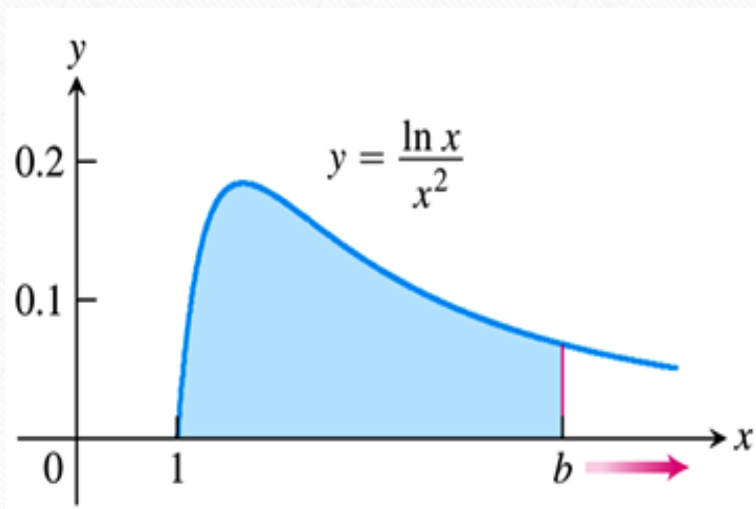


$$(c) \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$$



In each case, if the limit exists and is finite, we say that the improper integral **converges** and the limit is the **value** of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Ex1(p498) Is the area under the curve $y = \frac{\ln x}{x^2}$ from $x=1$ to $x=\infty$ finite? If so, what is its value?



Let $t \rightarrow \infty$

$$\lim_{t \rightarrow \infty} \int_1^t \frac{\ln x}{x^2} dx$$

$$= \lim_{t \rightarrow \infty} \int_1^t x^{-2} \ln x dx$$

$$\ln x \cdot x^{-2}$$

$$\frac{1}{x} \cdot \frac{-1}{x^2} = -x^{-1}$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{\ln x}{x} \right)_1^t + \int_1^t \frac{1}{x^2} dx$$

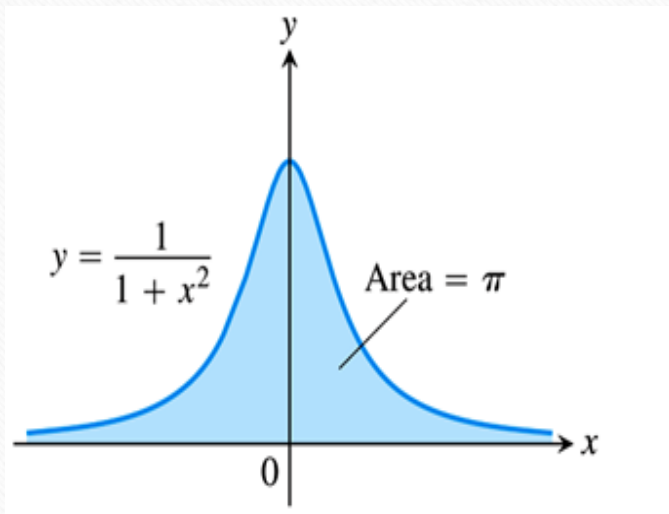
$$= \lim_{t \rightarrow \infty} \left(-\frac{\ln x}{x} \right)_1^t - \left(\frac{1}{x} \right)_1^t$$

$$= \lim_{t \rightarrow \infty} \left(\frac{-\ln x - 1}{x} \right)_1^t = \lim_{t \rightarrow \infty} \left(\frac{-\ln t - 1}{t} + \frac{0 + 1}{1} \right) = \lim_{t \rightarrow \infty} \left(-\frac{\ln t + 1}{t} + 1 \right) \left(-\frac{\infty}{\infty} \right)$$

$$= \lim_{t \rightarrow \infty} \left(\frac{-\frac{1}{t}}{1} + 1 \right) = 1$$

Ans: 收敛到 1 (converges to 1)

Ex2(p498) Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.



$$= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx = \lim_{s \rightarrow -\infty} \int_s^0 \frac{1}{1+x^2} dx + \lim_{t \rightarrow \infty} \int_0^t \frac{1}{1+x^2} dx$$

let $x = \tan \theta \quad dx = \sec^2 \theta d\theta$

$$\int_{-\frac{\pi}{2}}^0 \frac{1}{\sec^2 \theta} \sec^2 \theta d\theta = \theta \Big|_{-\frac{\pi}{2}}^0 = \frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sec^2 \theta} \sec^2 \theta d\theta = \theta \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$$\frac{\pi}{2} + \frac{\pi}{2} = \pi$$

$$\left| \begin{aligned} & \lim_{s \rightarrow -\infty} \int_s^0 \frac{1}{1+x^2} dx \\ &= \lim_{s \rightarrow -\infty} [\tan^{-1} x]_s^0 = \frac{\pi}{2} \\ & \lim_{t \rightarrow \infty} \int_0^t \frac{1}{1+x^2} dx \\ &= \lim_{s \rightarrow -\infty} [\tan^{-1} x]_0^t = \frac{\pi}{2} \end{aligned} \right.$$

Ex3(p499) For what values of p does the integral $\int_1^{\infty} \frac{1}{x^p} dx$ converges?

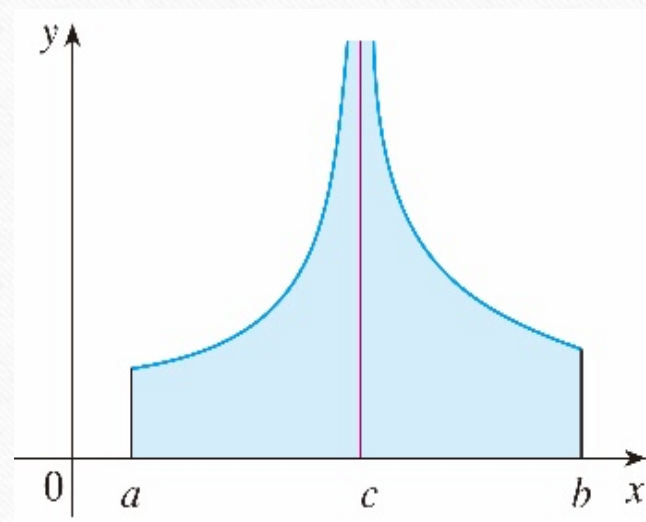
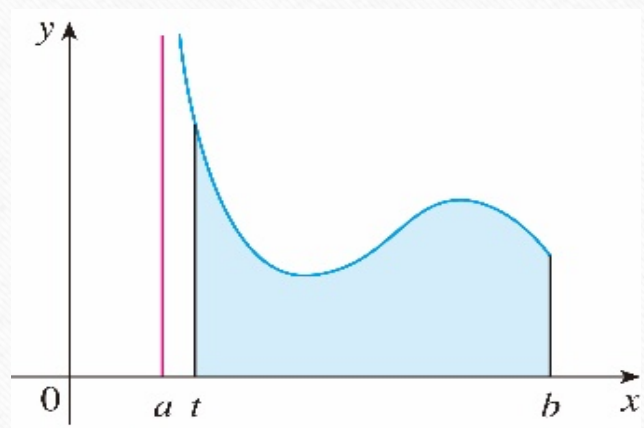
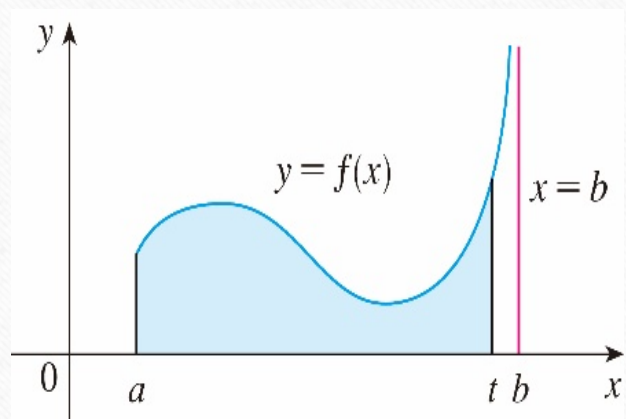
When the integral does converge, what is its value?

$$(1) \quad p \neq 1$$
$$\text{origin} = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \left(\frac{1}{-p+1} x^{-p+1} \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(\frac{1}{-p+1} t^{-p+1} - \frac{1}{-p+1} \right)$$

$$(2) \quad p = 1$$
$$\text{origin} = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} \left[\ln|x| \right]_1^t = \lim_{t \rightarrow \infty} \ln t - \ln 1 = \lim_{t \rightarrow \infty} \ln t = \infty$$

$$\begin{cases} p > 1, & \text{conv. to } \frac{1}{p-1} \\ p \leq 1, & \text{div.} \end{cases}$$

converges
diverges



Type II: Discontinuous Integrands

Definition of an Improper Integral of Type II

(a) If f is continuous on $[a, b)$ and is discontinuous at b , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

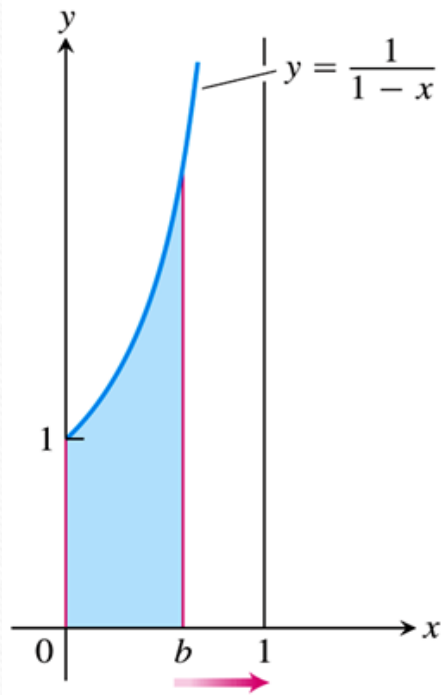
(b) If f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x) dx = \lim_{s \rightarrow a^+} \int_s^b f(x) dx$$

(c) If f has a discontinuity at c , where $a < c < b$, and continuous on

$[a, c) \cup (c, b]$, then $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

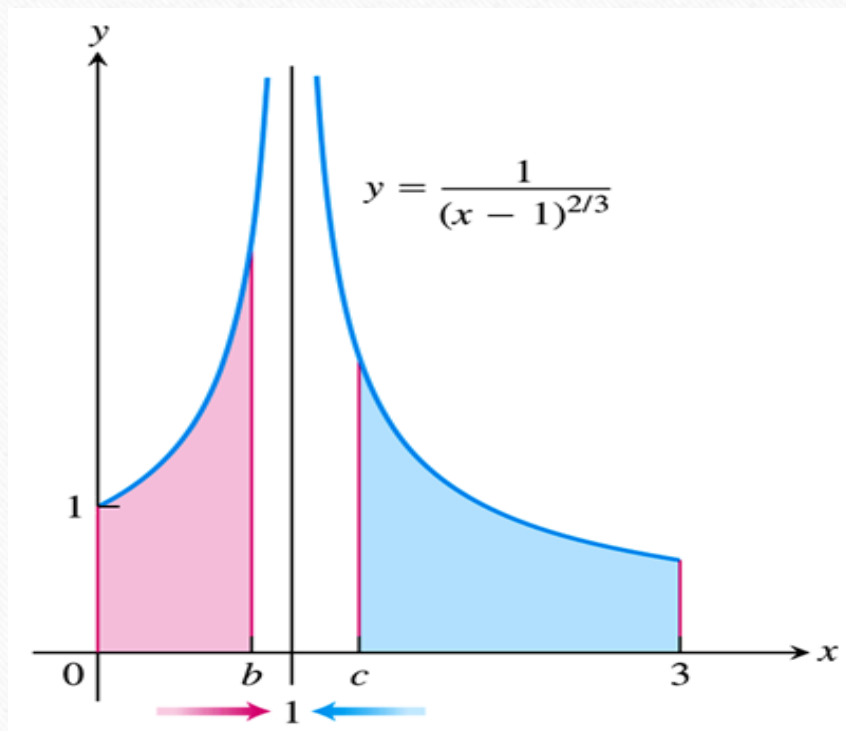
Ex4(p500) Investigate the convergence of $\int_0^1 \frac{1}{1-x} dx$.



$$= \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{1-x} dx = \lim_{t \rightarrow 1^-} -\ln|1-x| \Big|_0^t = \lim_{t \rightarrow 1^-} -\ln|1-t| = \infty$$

Divergent

Ex5(p501) Evaluate $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$.



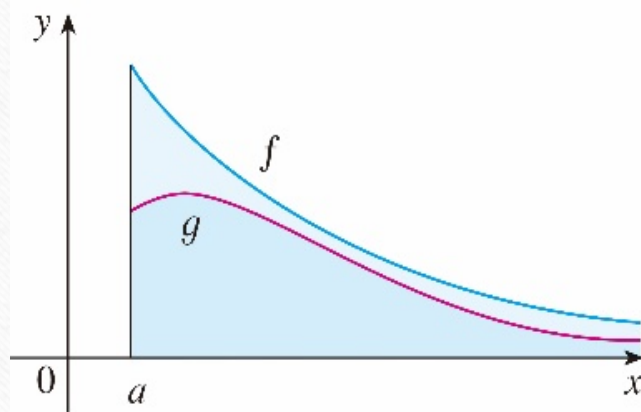
$$\begin{aligned}
 & \int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx \\
 &= \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{(x-1)^{\frac{2}{3}}} dx + \lim_{s \rightarrow 1^+} \int_s^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx \\
 &= \lim_{t \rightarrow 1^-} \left[-3(x-1)^{\frac{1}{3}} \right]_0^t + \lim_{s \rightarrow 1^+} \left[-3(x-1)^{\frac{1}{3}} \right]_s^3 \\
 &= \text{Divergent}
 \end{aligned}$$

直接比較審斂法

Theorem 2-Direct Comparison Theorem Suppose that f, g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

(a) If $\int_a^{\infty} f(x) dx$ is convergent, then $\int_a^{\infty} g(x) dx$ is convergent.

(b) If $\int_a^{\infty} g(x) dx$ is divergent, then $\int_a^{\infty} f(x) dx$ is divergent.



Ex7(p503) Determine the convergence of the integral (a) $\int_1^{\infty} \frac{\sin^2 x}{x^2} dx$.

$$\therefore 0 \leq \sin^2 x \leq 1, \quad \forall x \geq 1$$

$$\therefore 0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}, \quad \forall x \geq 1$$

$$\int_1^{\infty} \frac{1}{x^2} dx \text{ converges}$$

$$\therefore \int_1^{\infty} \frac{\sin^2 x}{x^2} dx \text{ converges, too}$$

極限比較審斂法

Theorem 3-Limit Comparison Test

If the positive functions f, g are continuous on $[a, \infty)$, and if

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty$, then $\int_a^\infty f(x) dx$ and $\int_a^\infty g(x) dx$ either both converge or both diverge.

Ex8(p504) Determine the convergence of the integral $\int_1^\infty \frac{dx}{1+x^2}$.

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{1+x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{x^2}{1+x^2} = 1$$

$\therefore \int_1^\infty \frac{1}{x^2} dx$ converges

By Limit Comparison Test

$\int_1^\infty \frac{1}{1+x^2} dx$ converges, too

HW8-7

- HW: 1,4,7,21,25,38,47

110

5. (10分) 計算以下瑕積分。(Evaluate the following improper integrals)

$$(1) \int_0^{\infty} x e^{-2x} dx$$

$$= \lim_{t \rightarrow \infty} \int_0^t x e^{-2x} dx$$

$$= \lim_{t \rightarrow \infty} \left(\frac{-x e^{-2x}}{2} - \frac{e^{-2x}}{4} \right) \Big|_0^t$$

$$= \lim_{t \rightarrow \infty} \left(\frac{-t e^{-2t}}{2} - \frac{e^{-2t}}{4} + \frac{1}{4} \right)$$

$$= \lim_{t \rightarrow \infty} \left(\frac{-t}{2e^{2t}} + \frac{1}{4} \right)$$

$$= \lim_{t \rightarrow \infty} \frac{-1}{4e^t} + \frac{1}{4} = \frac{1}{4}$$

$$(2) \int_0^1 \ln x dx$$

$$= \lim_{t \rightarrow 0^+} \int_t^1 \ln x dx = \lim_{t \rightarrow 0^+} (x \ln x - x) \Big|_t^1$$

$$\ln x \Big|_{\frac{1}{x}}^1 = \lim_{t \rightarrow 0^+} (-1 - t \ln t + t)$$

$$= \lim_{t \rightarrow 0^+} -t \ln t - 1$$

$$= \lim_{t \rightarrow 0^+} \frac{-\ln t}{\frac{1}{t}} - 1$$

$$= \lim_{t \rightarrow 0^+} \frac{-\frac{1}{t}}{-\frac{1}{t^2}} = \lim_{t \rightarrow 0^+} t = -1$$